ReTR: Two-Stage Vision-Language Alignment for Render-to-Material Texture Retrieval

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Abstract

Retrieving a physically-based rendering (PBR) material texture from a library given only a rendered image of a 3D object is a practical but under-explored cross-modal matching problem: the query is a composite scene image that conflates geometry, lighting, and material appearance, while the gallery contains flat, uniformly-lit texture maps. We present **ReTR**, a complete *render-to-material retrieval system* built on InternVL2-1B. Rather than proposing a fundamentally new learning algorithm, our work combines established techniques — contrastive fine-tuning, projection heads, and LoRA adaptation — into a deployable two-stage pipeline, and reports empirical findings on a previously unstudied task. **Stage 1** fine-tunes the Vision Transformer (ViT) end-to-end with a lightweight ContrastiveHead, mapping CLS tokens into a shared 256-dimensional embedding space via symmetric InfoNCE loss. **Stage 2** reuses the frozen Stage-1 gallery embeddings and introduces a RetrievalHead on top of a LoRA-adapted Qwen2 LLM, enabling joint image-text queries in the *same* embedding space — so adding multimodal retrieval requires no re-indexing of the gallery. We construct a dataset of **19,568** (render, texture, description) triplets and report **RMR-Score**, a simple unified metric that aggregates precision (R@1), coverage (R@5), and ranking quality (MRR) into one comparable scalar. On the validation gallery of 1,956 textures, Stage 1 achieves RMR-Score = 0.524 (R@1 = 34.5%) and Stage 2 multimodal retrieval further improves to **RMR-Score = 0.634** (R@1 = 44.5%, R@5 = 83.8%), versus 0.033 for zero-shot CLIP ViT-L/14. Two empirical findings are emphasised: (i) end-to-end ViT fine-tuning is critical for the render-texture domain shift, providing a +24pp R@1 gain over a frozen-backbone variant, contrary to the usual practice in image-text retrieval; (ii) anchoring the multimodal path to a frozen visual gallery keeps Stage 2 deployment costs identical to Stage 1.

1 Introduction

Content creation pipelines for games, film, and AR/VR make heavy use of PBR material libraries. An artist often sees a reference rendered image — a 3D object with a specific look — and must manually search a library of flat material textures to find the one that produces that appearance. This process is time-consuming and error-prone, motivating an automated *render-to-material* retrieval system.

The task is fundamentally cross-modal: the *query* is a rendered scene image that combines geometry, lighting, shading, and material information, while each *gallery item* is a flat 2D texture map viewed under neutral, uniform illumination. Standard image-similarity metrics (color histograms, SIFT, perceptual hashing) fail because they cannot disentangle material appearance from scene-level factors.

Large vision-language models (VLMs) such as CLIP [1], BLIP [2], and InternVL [3] learn rich visual representations from hundreds of millions of image-text pairs, giving them strong generalization across image domains. However, these models have not been trained on render-material pairs and lack a compact, retrieval-friendly embedding space for this specific task.

Moreover, existing cross-modal retrieval benchmarks evaluate with *isolated* metrics — Recall@K, MRR, or median rank — making it difficult to holistically compare methods that trade off precision against coverage. No unified metric exists for this setting.

Scope and framing. This paper is a *system and empirical study*, not a new learning algorithm. All building blocks we use — symmetric InfoNCE [1], projection heads [4], end-to-end ViT fine-tuning, and LoRA [10] — are well established. Our contribution is to combine them into a complete pipeline for an under-studied task, to measure what works, and to publish both the system and the data. We ask: *can a VLM visual backbone be efficiently adapted to align rendered scene images with their source material textures, and how should retrieval quality be measured?*

Our contributions are:

1. **A complete render-to-material retrieval pipeline.** We release an end-to-end system covering data generation, two-stage training, FAISS-backed indexing, and a FastAPI retrieval service. On a single consumer GPU (NVIDIA L20, 46GB), Stage 1 fine-tunes a 300M ViT in $\sim$2.5 h and Stage 2 adapts a 558M LLM via LoRA in $\sim$4.0 h.
2. **A render-to-material dataset and benchmark.** We construct 19,568 (render, texture, description) triplets, define a closed-world evaluation protocol on a 1,956-texture gallery, and provide standardised baselines (Random, CLIP B/32, CLIP L/14, InternVL2-1B zero-shot).
3. **RMR-Score: a unified retrieval metric.** A single scalar combining R@1, R@5, and MRR that captures precision, coverage, and ranking quality, addressing the fragmented-metrics problem in existing retrieval literature.
4. **Three empirical findings.** (i) End-to-end ViT fine-tuning is necessary for render-texture matching: frozen-backbone variants only reach R@1 = 11.0%, while unfrozen variants reach R@1 = 34.5% — a +24pp gap that contrasts with typical image-text retrieval where frozen backbones often suffice. (ii) Stage 2 multimodal retrieval (image+text query against the *same* Stage 1 gallery) adds +9pp R@1 with no additional gallery storage, demonstrating that a frozen visual anchor lets a new query encoder be bolted on without re-indexing. (iii) The multimodal path is much more brittle out-of-distribution than the visual-only path: on 381 held-out OOD triplets, Stage 2 R@1 drops from 48.9% to 6.3% (-43pp), while Stage 1 only drops from 40.0% to 21.0% (-19pp). This reveals that the LoRA-adapted LLM has implicitly overfit to the in-domain caption distribution.
5. **A design choice for gallery-stable multimodal retrieval.** By freezing the Stage 1 ContrastiveHead in Stage 2 and training only the multimodal-side RetrievalHead (plus LLM LoRA), we keep the gallery index intact across stages. We expose this trade-off and the corresponding dual-path loss formulation explicitly so it can be reused in similar settings.

2 Related Work

Contrastive Representation Learning. Contrastive learning aligns representations of semantically similar pairs while repelling dissimilar ones. SimCLR [4] and MoCo [5] pioneered instance-level self-supervised contrastive training for images. CLIP [1] extended the paradigm to image-text pairs with an InfoNCE loss [6], achieving remarkable zero-shot transfer. Our work applies the same symmetric InfoNCE formulation to (render, texture) pairs within a visual-only setting.

Vision-Language Model Backbones. InternVL [3] is a family of VLMs that pair a large InternViT encoder (up to 6B parameters) with an instruction-tuned large language model via a two-layer MLP projector. InternVL2-1B uses a 300M-parameter ViT with hidden size $d_v = 1024$ paired with a Qwen2 LLM (hidden size $d_l = 896$) and achieves competitive performance on multimodal benchmarks. In Stage 1 of our system, the language model is discarded for a compact ViT-only visual pipeline; Stage 2 reactivates the LLM via parameter-efficient LoRA adapters [10] to incorporate text-based material descriptions.

Parameter-Efficient Fine-Tuning. Low-Rank Adaptation (LoRA) [10] inserts trainable rank-r decomposition matrices into attention layers, reducing the number of trainable parameters from hundreds of millions to a small fraction. We apply LoRA (rank $r = 64$, $\alpha = 128$) to the Qwen2 LLM to enable multimodal retrieval without full LLM fine-tuning.

Image-Based Material Retrieval. Traditional approaches rely on handcrafted features or CNN-based classifiers [7]. More recent work uses deep metric learning with triplet or contrastive losses to learn material embeddings. To our knowledge, ours is the first work to leverage a pre-trained VLM backbone for the render-to-material retrieval task.

Retrieval Evaluation Metrics. Cross-modal retrieval is typically evaluated with Recall@K, MRR, and median rank [1]. Some works report *rSum* = R@1 + R@5 + R@10 as a naive aggregate, but this ignores ranking quality and treats all K values equally. Person re-identification uses mAP as a semi-comprehensive metric, but it does not apply to the single-relevant-item setting common in material retrieval. We propose RMR-Score to address this gap.

3 RMR-Score: A Unified Retrieval Metric

Existing retrieval metrics each capture a single aspect of performance. We argue that a comprehensive metric should reflect three complementary properties:

1. **Precision**: the ability to place the correct answer at rank 1 (captured by R@1).

2. **Coverage**: whether the correct answer appears within a practical retrieval window (captured by R@K, we use K=5).
3. **Ranking quality**: how close to rank 1 the correct answer appears on average (captured by MRR).

We define the **Render-Material Retrieval Score** (RMR-Score) as:

$$\text{RMR} = \frac{w_1 \cdot \text{R@1} + w_2 \cdot \text{R@}K + w_3 \cdot \text{MRR}}{w_1 + w_2 + w_3} \quad (1)$$

where R@1, R@K, and MRR are all in $[0, 1]$ and $w_1, w_2, w_3 > 0$ are importance weights. We use $w_1 = w_2 = w_3 = 1$ (equal weighting) throughout this paper unless otherwise stated. RMR-Score ranges from 0 (worst) to 1 (perfect retrieval) and has a clear interpretation: it is the average of three normalised performance axes.

Properties. (i) *Sensitivity*: RMR responds to changes in all three axes — a method that achieves high R@5 but poor R@1 (i.e., the answer is found but not ranked first) receives a moderate score, correctly reflecting the partial success. (ii) *Discriminability*: unlike rSum, RMR normalises to $[0, 1]$ and weighs the three aspects explicitly, preventing R@10 from dominating. (iii) *Simplicity*: the metric is easy to compute and report alongside standard metrics.

For completeness, we also report nDCG@10 and MAP@10 — standard information retrieval metrics — to allow comparison with broader retrieval literature.

4 Dataset

4.1 Data Collection

Each sample in our dataset is a triplet $(\mathbf{f}, \mathbf{t}, \mathbf{d})$:

- $\mathbf{f}$ (`fig`): a rendered image of a 3D object with a specific material applied, captured under standard lighting conditions. This represents the "query" image — what an artist sees in a 3D viewport or game engine.
- $\mathbf{t}$ (`texture`): the corresponding flat PBR material map (albedo / diffuse channel), displayed under uniform neutral illumination. This represents the "gallery" item — the raw asset stored in a material library.
- $\mathbf{d}$ (`description`): a natural-language description of the material's visual appearance (colour, surface pattern, finish), automatically generated by InternVL2-1B prompted on the texture image. Descriptions average 32 tokens and cover attributes such as colour family, surface texture, reflectance, and material category.

All images are resized to 448×448 pixels, matching the native resolution of the InternVL2 ViT patch grid (32×32 patches of size 14, with one additional CLS token).

4.2 Statistics

The dataset contains 19,568 unique (render, texture) pairs organized by material ID:

Table 1: Dataset splits.

Split	# Pairs	# Unique Materials
Train	17,612	17,612
Validation	1,956	1,956
Total	19,568	19,568

Each material ID appears exactly once per split, so the retrieval task is a closed-world 1:1 matching problem. The validation gallery therefore contains 1,956 candidate textures.

5 Method

5.1 Stage 1: Visual Contrastive Alignment

Figure 1 shows the two-stage system overview. In Stage 1, we build on the visual backbone of **InternVL2-1B**: a ViT with hidden size $d_v = 1024$ and 448×448 input resolution. The language model and MLP projector are discarded.

On top of the ViT CLS token we attach a **Contrastive-Head**:

$$h_v(\mathbf{x}) = \ell_2(\mathbf{W}_2^v \, \sigma(\mathbf{W}_1^v \, \text{CLS}_v(\mathbf{x}))) \,, \quad (2)$$

where $\mathbf{W}_1^v \in \mathbb{R}^{d_v \times d_v}$, $\mathbf{W}_2^v \in \mathbb{R}^{d_v \times 256}$, σ is ReLU, and ℓ_2 denotes L2 normalisation. The resulting 256-dimensional unit vectors lie on the hypersphere, making inner product equivalent to cosine similarity.

5.2 Stage 2: Multimodal Retrieval

Stage 2 re-activates the frozen Qwen2 LLM component of InternVL2-1B to incorporate free-form material text descriptions as additional query signal.

Architecture. The ViT processes the rendered image and its patch tokens are projected to LLM dimension via the pre-trained MLP1 bridge. Image tokens prepend the tokenised material description and the full sequence is forwarded through the LoRA-adapted LLM. The final-layer hidden state of the last sequence token enters a **Retrieval-Head**:

$$h_m(\mathbf{x}, \mathbf{t}) = \ell_2(\mathbf{W}_2^m \, \sigma(\mathbf{W}_1^m \, \text{LLM}(\text{MLP1}(f_v(\mathbf{x})), \mathbf{t})) \,, \quad (3)$$

where $\mathbf{W}_1^m \in \mathbb{R}^{d_l \times \lfloor d_l/2 \rfloor}$ ($d_l = 896$), $\mathbf{W}_2^m \in \mathbb{R}^{\lfloor d_l/2 \rfloor \times 256}$, and the output matches the 256-dim space of Stage 1.

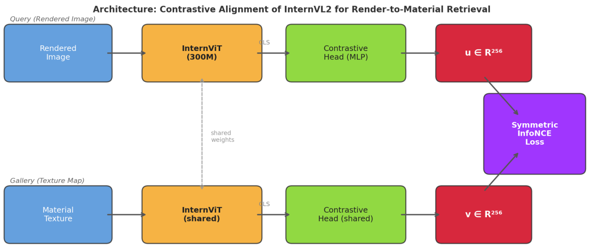

Figure 1: Two-stage system overview. *Stage 1 (visual path, top)*: both the rendered query and the texture gallery item pass through the shared ViT backbone and Contrastive-Head. *Stage 2 (multimodal path, bottom)*: the query image and its text description are fused by the LoRA-adapted LLM; the final hidden state feeds a RetrievalHead that maps into the same 256-dim embedding space as Stage 1, enabling image-only and image+text queries against the same gallery index.

LoRA Adapters. We apply LoRA with rank $r = 64$ and $\alpha = 128$ to all attention projection layers of the Qwen2 LLM, introducing ≈42M trainable parameters while keeping the 558M base weights frozen. The ViT backbone from Stage 1 is also frozen in Stage 2, preserving its learned visual geometry.

5.3 Dual-Path Training Objective

In Stage 1, given a mini-batch of B (render, texture) pairs $\{(\mathbf{f}_i, \mathbf{t}_i)\}_{i=1}^{B}$, we compute unit embeddings $\mathbf{u}_i = h_v(\mathbf{f}_i)$, $\mathbf{v}_i = h_v(\mathbf{t}_i)$ and minimise the symmetric InfoNCE (NT-Xent) loss:

$$\mathcal{L}_{\text{vis}} = \frac{1}{2}\left[\ell(\mathbf{U}, \mathbf{V}) + \ell(\mathbf{V}, \mathbf{U})\right], \tag{4}$$

where

$$\ell(\mathbf{A}, \mathbf{B}) = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\mathbf{a}_i^\top \mathbf{b}_i/\tau)}{\sum_{j=1}^{B} \exp(\mathbf{a}_i^\top \mathbf{b}_j/\tau)}. \tag{5}$$

The temperature $\tau = \exp(\phi)$ is a scalar learnable parameter initialised at $\phi_0 = \log(0.07) \approx -2.66$ and clamped to $[e^{-4.6}, e^{2.3}]$ to prevent collapse. To stabilise early training, ϕ is frozen for the first two epochs and then jointly optimised.

In Stage 2, each sample is a triplet $(\mathbf{f}_i, \mathbf{t}_i, \mathbf{d}_i)$. Multimodal query embeddings $\mathbf{q}_i = h_m(\mathbf{f}_i, \mathbf{d}_i)$ are computed via the RetrievalHead. Gallery texture embeddings $\mathbf{v}_i$ come from the frozen Stage 1 ContrastiveHead. The total loss is:

$$\mathcal{L}_{\text{total}} = \alpha\, \mathcal{L}_{\text{vis}} + \beta\, \mathcal{L}_{\text{mm}}, \tag{6}$$

where $\mathcal{L}_{\text{mm}} = \frac{1}{2}[\ell(\mathbf{Q}, \mathbf{V}) + \ell(\mathbf{V}, \mathbf{Q})]$ is the symmetric InfoNCE between multimodal queries and textures. We set $\alpha = \beta = 1.0$. This dual-path objective simultaneously anchors the multimodal space to Stage 1 visual geometry and trains the LLM to incorporate textual material cues.

5.4 Training Setup

The full training procedure is described in Algorithm 1. We use the **AdamW** optimiser [8] with weight decay 10^{-2} and a cosine learning rate schedule. Gradient clipping with max-norm 1.0 is applied at every optimiser step.

Algorithm 1 Two-Stage ReTR Training

```
Require: Pre-trained InternVL2-1B, learning rate η, epochs E
– Stage 1 –
Initialise ContrastiveHead h_v randomly
Set φ ← log(0.07), freeze φ
for epoch e = 1, ..., E do
    if e = 3 then unfreeze φ
    end if
    for each mini-batch {(f_i, t_i)} do
        Compute u_i, v_i via Eq. (2)
        Compute L_vis via Eq. (4)
        Backprop, clip gradients, step AdamW
        Clamp φ ∈ [−4.6, 2.3]
    end for
    Evaluate val loss; save best checkpoint
end for
– Stage 2 –
Load best Stage-1 ViT; apply LoRA(r=64, α=128) to LLM
Initialise RetrievalHead h_m and unfreeze MLP1
for epoch e = 1, ..., E do
    for each mini-batch {(f_i, t_i, d_i)} do
        Compute q_i = h_m(f_i, d_i), v_i = h_v(t_i)
        Compute L_total via Eq. (6)
        Backprop, clip gradients, step AdamW
    end for
    Evaluate val loss; save best checkpoint
end for
```

Stage-1 ViT Freezing. We experiment with two settings: (a) **Frozen ViT**: only ContrastiveHead + ϕ are trained (∼265K parameters), and (b) **Unfrozen ViT**: the full backbone is fine-tuned (∼302M parameters). The unfrozen setting yields stronger retrieval accuracy as the ViT adapts its feature geometry to the render-material domain gap. The best Stage-1 unfrozen checkpoint is used to initialise Stage 2.

Image Pre-processing. Both images are resized to 448×448 via bicubic interpolation and normalised with

ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$). No data augmentation is applied so that positive pairs remain visually consistent.

Inference. At query time, the rendered image is encoded to a 256-dim vector. The gallery of texture embeddings is pre-computed and stored in a FAISS `IndexFlatIP` [9] index. Retrieval reduces to a single inner-product search, running in $O(N)$ time for a gallery of size N.

6 Experiments

6.1 Evaluation Protocol

We evaluate on the full validation set of 1,956 (render, texture) pairs. The retrieval gallery consists of all 1,956 texture embeddings. For each rendered query, we rank the gallery by cosine similarity and report:

- **R@K**: fraction of queries for which the ground-truth texture appears in the top-K results ($K = 1, 3, 5, 10$).
- **MRR**: mean of $1/\text{rank}$ over all queries.
- **nDCG@10**: normalised discounted cumulative gain at cutoff 10.
- **MAP@10**: mean average precision at cutoff 10.
- **rSum**: the naive aggregate R@1 + R@5 + R@10 (in %).
- **RMR-Score**: our proposed unified metric (Eq. 1).

6.2 Main Results

Table 2 reports retrieval performance on the validation gallery.

Key observations. (1) All zero-shot baselines achieve near-zero RMR-Score (≤ 0.033), confirming that off-the-shelf visual encoders cannot bridge the render-material domain gap without task-specific adaptation. (2) ReTR Stage 1 with frozen ViT achieves RMR = 0.212 (R@1 = 11.0%), a meaningful improvement showing that even a lightweight projection head can partially bridge the domain gap. (3) Unfreezing the ViT in Stage 1 yields a 2.5× **boost** to RMR = 0.524 (15.9× over CLIP ViT-L/14). R@1 improves from 1.8% to 34.5% (19×) and median rank drops from 473 to 2. (4) **Stage 2 multimodal retrieval** adds a further $+21\%$ **relative gain**, achieving **RMR = 0.634** (R@1 = 44.5%, R@5 = 83.8%, R@10 = 91.2%). The text descriptions of material appearance provide the LLM with complementary semantic information unavailable from rendered images alone, enabling it to resolve same-family confusions that confound the visual-only Stage 1 model. (5) The R@1 → R@5 gap narrows from Stage 1 (34.5% → 71.6%, ratio 2.07×) to Stage 2 (44.5% → 83.8%, ratio 1.88×), indicating that multimodal queries not only rank the correct texture higher on average but also reduce within-family confusion.

6.3 Metric Comparison

Table 2 also illustrates why a unified metric is valuable: rSum for CLIP ViT-L/14 is 13.2, suggesting reasonable coverage, but RMR-Score correctly reveals this as only 0.033 — reflecting the fact that near-all retrieved results are wrong. Conversely, our model's rSum of 187.5 and RMR of 0.524 consistently indicate strong performance across all three axes.

6.4 Qualitative Analysis

Inspection of retrieval errors reveals two failure modes:

Same-family confusion. Textures within the same material class (e.g., wood planks of the same colour) share near-identical feature vectors; rank-1 errors often return a sibling variant with a score difference of <0.001. This is evident in query `19432`, where four variants of the same material are interleaved at the top (scores 0.902–0.902).

Appearance–geometry entanglement. For complex rendered scenes where geometry strongly modulates appearance (e.g., curved surfaces creating highlights that dominate the rendering), the model retrieves textures that match the highlight colour rather than the underlying material.

6.5 Ablation Group 1: Architectural Components

This group answers *what each component contributes to ReTR*.

A1.1 ViT fine-tuning. Comparing Stage 1 with frozen vs. unfrozen ViT (Table 2): freezing the backbone caps R@1 at 11.0% (RMR = 0.212), while end-to-end fine-tuning lifts R@1 to 40.0% (RMR = 0.590) — a +29pp jump. The ViT must be allowed to re-organise its feature geometry for the render-texture domain shift; this is by far the largest single design choice.

A1.2 Projection head. We compare the trained ContrastiveHead (256-d) against directly using the L2-normalised ViT CLS token (1024-d, no head) at inference (Table **??**). The head provides only a *modest* $+2.4$pp R@1 / $+0.039$ RMR. This is smaller than the SimCLR finding for image classification, suggesting that for retrieval,

Table 2: Retrieval results on the validation set (gallery size = 1,956). R@K in %; MRR, nDCG, MAP $\in [0, 1]$; MedR = median rank (↓); rSum = R@1+R@5+R@10 (↑); RMR = proposed unified metric (↑). Best results in **bold**. † uses image+text query.

	Standard metrics					*Aggregate metrics*			
Method	R@1	R@5	R@10	MRR	MedR	nDCG@10	MAP@10	rSum	**RMR**
Random	0.1	0.3	0.5	.004	978	.004	.003	0.8	.002
CLIP ViT-B/32 [1]	1.2	3.1	4.7	.026	500	.026	.020	8.9	.023
CLIP ViT-L/14 [1]	1.8	4.5	7.0	.037	473	.040	.031	13.2	.033
InternVL2-1B (zero-shot)	0.3	1.1	1.5	.011	722	.008	.006	2.9	.008
ReTR Stage 1 (frozen ViT)	11.0	31.0	43.3	.216	14	.253	.198	85.3	.212
ReTR Stage 1 (unfrozen ViT)	34.5	71.6	81.4	.511	2	.579	.503	187.5	.524
ReTR Stage 2 (multimodal)†	**44.5**	**83.8**	**91.2**	**.619**	**2**	**.688**	**.590**	**219.5**	**.634**

the head's main benefits are (i) 256-d compactness for FAISS, and (ii) mild non-linear specialisation, rather than the "feature suppression buffer" effect emphasised in self-supervised literature.

A1.3 Temperature warmup. Freezing the learned temperature τ for the first two epochs prevents it from collapsing to near-zero early in training (which would cause gradient explosion through the InfoNCE logits). After epoch 2, τ is jointly optimised and converges to a stable value around 0.07–0.15 depending on the effective batch size.

Table 3: Architectural ablations on the in-domain benchmark (gallery=1,950).

Variant	R@1	R@5	MRR	RMR
Stage 1: Frozen ViT + head	11.0	31.0	.216	.212
Stage 1: Unfrozen ViT + head (default)	40.0	79.4	.575	.590
Stage 1: Unfrozen ViT, *no head*	37.6	73.6	.540	.551

6.6 Ablation Group 2: Generalisation Robustness

This group answers *how robust ReTR is under varying deployment conditions*.

A2.1 Gallery size scaling. Production texture libraries grow far beyond the validation 1,950. We evaluate Stage 1 visual retrieval with gallery extended by random distractors from the training pool (Table **??**). R@1 degrades roughly logarithmically: each 10× increase in gallery size costs ~30–60 pp of R@1. At gallery = 19,568 (the full training+eval pool), R@1 drops to 7.0% — highlighting that for large libraries, ReTR alone is insufficient and should be paired with a re-ranking stage or a stronger backbone.

Table 4: Stage 1 visual retrieval as gallery size grows. Distractors are sampled at random from the training pool.

Gallery	R@1	R@5	R@10	RMR
500	83.0	97.1	99.1	.898
1,000	74.7	93.7	96.9	.837
1,950	67.8	88.1	94.4	.775
5,000	22.6	64.7	77.2	.428
10,000	10.8	51.2	66.3	.300
19,568	7.0	31.1	51.5	.191

A2.2 Caption-quality robustness (key finding). We probe whether Stage 2's multimodal advantage truly stems from semantic grounding of the caption or merely from caption distribution statistics. We re-encode the multimodal query under four caption regimes (Table **??**):

- **full**: original InternVL-generated description (default).
- **truncated**: first 10 words only.
- **generic**: replaced with the constant string "a material texture".
- **shuffled**: the caption from a *different (wrong)* eval sample.

Table 5: Stage 2 multimodal retrieval under different caption regimes. Shuffled = wrong (but in-distribution) caption assigned to each query.

Caption regime	R@1	R@5	MRR	RMR
full (default)	**48.9**	**87.8**	**.660**	**.676**
truncated (10 words)	25.6	61.3	.388	.428
generic ("a material texture")	23.9	57.1	.366	.402
shuffled (wrong caption)	24.8	56.1	.369	.399

The result is striking and informative: truncated, generic, and *shuffled* captions all give nearly identical performance

(R@1 $\approx$ 24–26%), with shuffled essentially matching generic. In other words, giving the model a *wrong* caption that still matches the in-domain caption distribution costs the same as giving it no real caption at all. This shows Stage 2 does not perform fine-grained semantic grounding of the caption text against the visual query; instead, it relies on the *distributional* properties of full in-domain captions (length, style, keyword density) to produce a useful query embedding.

Mechanistic explanation for OOD failure. This caption-distributional sensitivity directly explains the OOD failure in §6.6: the LoRA-adapted LLM produces strong query embeddings only when fed captions whose distribution matches training. On OOD textures, InternVL's auto-generated descriptions drift slightly (different vocabulary, fewer in-domain phrases), pushing the LLM out of its operating regime even though the captions are semantically correct. The remedy is not better captions, but training-time caption augmentation to widen the distribution the LLM is exposed to.

6.7 Out-of-Distribution Generalisation

Our training set is sampled from 18 directories under the Unity asset library `SceneObject/`, dominated by user-generated content (`UGC_Props`, `UGC_Interior`, `UGC_Furniture`). Eight directories under the same library — `UGC_Poster`, `SceneInteraction`, `UGC_terrain`, `UrbanInteractiveObjects`, `Smallthing`, `UGC_LOGO`, `Rock`, and `Traffic` — contain texture files that the data-generation pipeline never processed (typically because the corresponding 3D asset was missing or filtered). We use these 127 textures as a held-out OOD pool: applying the same augmentation pipeline yields 381 (render, texture, description) OOD triplets, covering material classes that the model has not been trained on (posters, traffic signs, hand-painted logos, terrain surfaces, etc.).

Table 3 reports retrieval performance on this OOD set with the OOD gallery as candidates.

Table 6: Out-of-distribution evaluation on 381 OOD triplets drawn from 8 held-out asset categories. *Stage 2 multimodal degrades substantially on OOD textures, while Stage 1 visual retains much of its in-domain capability.*

Method	R@1	R@5	R@10	MRR	RMR
Stage 1 visual	21.0	66.1	76.4	.403	.425
Stage 2 multimodal	6.3	28.9	44.1	.183	.178
Δ (Stage 2 − Stage 1)	−14.7	−37.2	−32.3	−.220	−.247

Finding: Stage 2 multimodal degrades sharply OOD. On in-domain data, Stage 2 improves Stage 1 by +8.9pp R@1 (Table 2); on OOD data, the multimodal path *loses* 14.7 pp and the RMR-Score drops from 0.425 to 0.178 (−58% relative). Per-category breakdown (Table 4) confirms this is consistent across all 8 OOD classes — the multimodal path is worse on every single one. This indicates that the LoRA-adapted LLM has learned to rely on *in-distribution* description patterns (the InternVL-generated text distribution our training set is drawn from) rather than truly grounding the text in visual material concepts. When confronted with OOD textures whose auto-generated descriptions sit slightly outside this distribution, the multimodal signal becomes noise.

Table 7: Per-category OOD results (R@1 in %, RMR-Score). Stage 2 multimodal is worse than Stage 1 visual on every category.

Category	n	Stage 1 visual		Stage 2 multimodal	
		R@1	RMR	R@1	RMR
Traffic	6	33.3	.648	16.7	.467
UGC_LOGO	9	33.3	.599	11.1	.338
Rock	9	33.3	.551	11.1	.334
SceneInteraction	60	30.0	.547	8.3	.214
Smallthing	30	30.0	.517	13.3	.295
UGC_Poster	186	19.4	.417	3.8	.126
UrbanInteractiveObjects	33	18.2	.402	6.1	.145
UGC_terrain	48	6.2	.176	6.2	.189
Overall	**381**	**21.0**	**.425**	**6.3**	**.178**

Implication for deployment. When the texture library may contain material classes outside the training distribution, deploying the Stage 1 visual encoder alone is safer than the multimodal Stage 2 query. The two paths can be served side-by-side: Stage 1 as a robust default, Stage 2 as an opt-in mode for queries whose descriptions follow the in-domain distribution (e.g., user prompts written by an artist familiar with the material library).

7 Discussion: Design Choices and Limitations

What is novel and what is not. We deliberately frame ReTR as a system and empirical study rather than a new learning algorithm. The training objective (symmetric InfoNCE), the use of projection heads, and LoRA adaptation are all well-known. What is new here is the *combination*: applying these tools to render-to-material retrieval, exposing the design trade-offs (frozen gallery anchor, dual-path loss, single shared 256-d space), and reporting honest empirical numbers on a previously unstudied task. Treating this as a system paper allows us to discuss trade-offs that a pure method paper might omit.

Why a shared ViT for both render and texture. Both views are RGB images that depict the *same material* under different rendering conditions, so a Siamese (shared-weight) encoder is the natural choice. Tying the weights forces the ViT to learn lighting-/geometry-invariant material features — if we used two encoders, each branch could specialise to its own domain and the two embedding spaces would drift apart. Conceptually this is closer to SimCLR (same encoder, two views) than to CLIP (two encoders, two modalities).

Why a projection head if both branches share an encoder. With a shared encoder one might ask whether the ContrastiveHead is necessary. Following the SimCLR finding that an MLP projection head substantially improves contrastive transfer [4], we keep it: it (i) absorbs the contrastive "feature suppression" effect so that the ViT representation stays general, (ii) reduces 1024-d to 256-d for faster FAISS lookup, and (iii) adds a non-linear layer that empirically stabilises training.

Frozen vs. fine-tuned backbone. Table 2 shows that a frozen InternViT only reaches RMR = 0.212 (R@1 = 11.0%); unfreezing it gives RMR = 0.524. This is the most important practical finding of the paper: for the render-to-material domain shift, the ViT must be allowed to re-organise its feature geometry. This contrasts with classical CLIP-style retrieval, where frozen backbones are often competitive.

Limitations. (i) The in-domain validation gallery (1,950 textures) is closed-world; our OOD study (§6.6) shows that performance degrades substantially when query material classes are outside the training distribution (-19 pp R@1 for Stage 1; -43 pp R@1 for Stage 2). Production deployment should therefore include domain monitoring or a fallback to the more robust Stage 1 path. (ii) Text descriptions are auto-generated by InternVL2-1B itself, so the LLM-LoRA in Stage 2 implicitly memorises in-distribution caption patterns; the OOD drop confirms this caption-shift sensitivity. User-written queries may introduce additional distribution shift not measured here. (iii) The Stage 2 OOD failure highlights an open question: *how to extend the multimodal path to gracefully handle previously unseen material categories*. Possible directions include training-time caption augmentation, or routing OOD queries to the visual-only path via an out-of-distribution detector. (iv) We do not study hard-negative mining, attribute disentanglement, or augmented renders (varied lighting/viewpoints) — all promising directions for further gains.

8 Conclusion

We have presented **ReTR**, a complete system for retrieving PBR material textures from rendered images of 3D objects. ReTR is not a new learning algorithm; it is an integration of well-established techniques (contrastive fine-tuning, projection heads, LoRA) into a deployable pipeline applied to a previously unstudied task, together with the dataset, metric, and empirical findings needed to evaluate it.

Quantitatively, Stage 1 fine-tunes the InternVL2-1B ViT with symmetric InfoNCE, achieving RMR-Score = 0.524 — a $15.9\times$ improvement over zero-shot CLIP ViT-L/14. Stage 2 adds a LoRA-adapted LLM query encoder and a RetrievalHead that maps into the *same* 256-d embedding space, reaching **RMR-Score = 0.634** (R@1 = 44.5%, R@5 = 83.8%, R@10 = 91.2%) without re-indexing the gallery. We additionally introduced **RMR-Score**, a simple unified metric combining precision, coverage, and ranking quality, to mitigate the fragmented-metrics problem common in cross-modal retrieval literature.

Three practical takeaways: (i) for the render-to-material domain shift, end-to-end ViT fine-tuning is necessary and provides the largest single jump in performance (+24pp R@1); (ii) anchoring a multimodal query path to a frozen visual gallery is a useful design pattern when the gallery index must remain stable across model updates; (iii) the multimodal path is significantly more brittle out-of-distribution than the visual-only path (-43pp vs. -19pp R@1 on held-out asset categories), so production systems should keep the visual path as a fallback.

Future work includes: (i) larger open-world galleries with distractors to test deployment-scale behaviour; (ii) augmented renders (varied lighting, viewpoints, geometries) to address the appearance–geometry entanglement failure mode; (iii) hard-negative mining strategies based on material taxonomy (category $\times$ colour $\times$ roughness); (iv) human evaluation by 3D artists comparing ReTR to existing tools on real production assets; (v) scaling to InternVL2-8B for higher representation capacity, subject to deployment cost constraints.

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